Add Google Analytics to R Shiny App

1. Create a work-associated Google Account

Sign in to Google Analytics with your Allen Institute account. In Admin, create the 'property' that will contain the app data (Admin > Create button > Property). Give it a reasonable name (e.g., the app name) and then fill out the business details.

2. Create a new GA4 property

Sign in to Google Analytics with your Allen Institute account. In Admin, create the 'property' that will contain the app data (Admin > Create button > Property). Give it a reasonable name (e.g., the app name) and then fill out the business details.

3. Create a web data stream

This is done in one of two ways:

1. As the fourth step of property creation, follow the steps to 'Start collecting data.' (platform > Web)
2. Under Admin > Data collection and modification > Data streams, create a Web stream (Add Stream > Web).

Enter the deployed Shiny app URL and an app-specific stream name.

4. Copy the measurement ID

Open the web stream and copy its Measurement ID, which has the form G-XXXXXXXXXX. This step may not be needed as the html code for the Google tag (example shown below, but with G-XXXXXXXXXX included) may pop up automatically upon creation of the web data stream.

5. Add the Google tag to your app

First, create a file called "google-analytics.html" with the following content, replacing G-XXXXXXXXXX with the measurement ID above.

```
<!-- Google tag (gtag.js) -->
<script async src="https://www.googletagmanager.com/gtag/js?id=G-XXXXXXXXXX"></script>
<script>
  window.dataLayer = window.dataLayer || [];
  function gtag(){dataLayer.push(arguments);}
  gtag('js', new Date());

  gtag('config', 'G-XXXXXXXXXX');
</script>
```

Next, add a call-out to this file in the app UI (either in app.R or ui.R, depending on app setup).

```
ui <- fluidPage(
  tags$head(
    includeHTML("google-analytics.html")
  ),
  # Existing UI content
)
```

If you are deploying automatically from Code Ocean, add "google-analytics.html" to your list of deployed app files, either by updating the deploy.yml file OR by running `rsconnect::writeManifest()` as needed, depending on how the app was initially deployed.

6. Deploy the app

Deploy or redeploy the app normally. The Google tag runs in the user's browser, so no additional R package is required.

7. Verify data collection

Open the deployed app in a new browser window. In Google Analytics, check Reports > Realtime and confirm that an active user or page_view appears. Alternatively, upon completion of set-up you may be given an option to "Test installation". Data may not appear in standard reports for 12-48 hours, although testing should work upon completed deployment.

Notes

- Basic setup records page views. Tracking actions such as plot downloads, data downloads, or shared views requires additional GA4 event code.
- Use app-specific streams and confirm Allen Institute privacy requirements before collecting more than anonymous aggregate usage.

References

- Posit: Add Google Analytics: https://shiny.posit.co/r/articles/build/google-analytics/
- Google: About the gtag.js snippet: https://support.google.com/analytics/answer/10271001?hl=en